

1. Suppose there are 200 observations and we perform **10-fold cross-validation**.

For each fold, KNN is trained on 180 observations and evaluated on 20.

Question: Across the entire 10-fold CV procedure:

1. How many times is each observation used for validation?
2. How many times is each observation used for training?
3. How many models are trained?

1 - Once

2 - 9 times

3 - 10 models, one per fold

2. You perform 5-fold CV and get

(k)	Mean CV Accuracy
1	91%
3	93%
5	93%
7	92%

Suppose the standard deviations of the five fold accuracies are:

(k)	Mean	SD
1	91%	1%
3	93%	2%
5	93%	0.5%
7	92%	1%

Which k would you prefer between $k = 3$ and $k = 5$, and why?

$k=5$ $\rightarrow$ both $k=3$ and $k=5$ have same cv mean.
A lower standard dev. gives a more consistent performance across folds.

3. Suppose:

(k)	Test Accuracy
5	91.2%
7	91.3%
9	91.2%
11	90.9%

Would you necessarily choose $k = 7$? Why/Why not?

no $\rightarrow$ highest test acc does not best
k should be chosen w/ cross validation

4. Suppose you are comparing two choices of k

$k = 3$

Fold	Accuracy
1	95%
2	95%
3	95%
4	75%
5	75%

$k = 7$

Fold	Accuracy
1	89%
2	90%
3	91%
4	89%
5	90%

Which model appears more stable and why?

$k = 7 \rightarrow$ less deviation

5. Suppose cross validation gives:

(k)	CV Accuracy
1	88%
3	91%
5	94%
7	93%
9	92%

You choose $k = 5$ and evaluate the model on the test set and get 90% accuracy. Your colleague says “Since $k = 7$ might have done better on the test set, let's try $k = 7$ and see.”

Is this a good or bad idea? Why?

6. An airport security system uses ML to identify potentially dangerous passengers.

Suppose the system has:

- 100 dangerous passengers
- 10,000 safe passengers

The system catches 95 of the 100 dangerous passengers but incorrectly flags 500 safe passengers.

- What is recall?
- What is precision?
- What metric should be prioritized?

7. Imagine you're building a search engine.

A user searches: "machine learning textbooks"

The search engine returns 20 results. Only 15 are actually relevant.

- Which metric is directly measuring how useful those 20 returned results are?

8. Two models are being considered for detecting a disease:

Model	Precision	Recall
A	95%	60%
B	80%	95%

The disease is serious, and the consequences of **missing a sick patient** are much greater than the consequences of sending a healthy patient for additional testing.

Which model would you choose and why?